

SSL CERTIFICATE EXPIRY WATCHER

1. Project Objective

The objective of this project is to design and develop an automated SSL Certificate Monitoring System that continuously tracks SSL certificates of public domains and proactively identifies expiry risks.

The system aims to:

- Monitor SSL certificates automatically
- Detect expiry risks before outages occur
- Generate AI-powered renewal recommendations
- Send real-time notifications
- Store monitoring history
- Provide operational dashboards and reports
- Reduce manual SSL management effort

2. Problem Statement

Organizations often manage multiple websites and services secured by SSL/TLS certificates.

Manual certificate tracking can result in:

- Unexpected certificate expiration
- Website downtime
- Security vulnerabilities
- Loss of customer trust
- Increased operational workload

An automated monitoring solution is required to ensure certificate validity and timely renewal actions.

3. Proposed Solution

The SSL Certificate Expiry Watcher automatically scans configured domains, retrieves SSL certificate information, evaluates expiry risks, and generates renewal actions using Artificial Intelligence.

The platform also provides:

- Real-time alert notifications
- Historical monitoring records
- Priority-based domain ranking
- Dashboard visualization
- Automated report generation

4. Project Modules

Module	Responsibility
scanner.py	Retrieve SSL certificate metadata
engine.py	Calculate risk levels and incidents
llm.py	Generate AI-based renewal recommendations
database.py	Store monitoring history in SQLite
discord_alert.py	Send Discord notifications
ranker.py	Prioritize domains by urgency
report.py	Generate CSV and Markdown reports
dashboard.py	Provide Flask web dashboard

5. System Architecture

domains.csv

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SSL Scanner

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Risk Classification Engine

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Incident Detection Engine

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Ollama (Llama3)

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SQLite Database

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Domain Ranking Engine

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CSV & Markdown Reports

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Dashboard

6. Project Modules

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7. Functional Requirements

7.1 SSL Monitoring

The system shall:

- Read domains from CSV input
- Establish SSL connections
- Retrieve certificate details
- Extract expiration date
- Calculate remaining validity period

7.2 Risk Classification

The system shall classify domains as:

Status	Condition
HEALTHY	More than 30 days remaining
WARNING	7–30 days remaining
BROKEN	Less than 7 days remaining
ERROR	Certificate retrieval failed

7.3 AI Task Generation

The AI module shall generate:

- Recommended renewal steps
- Priority levels
- Suggested ownership
- Action plans
- Risk summaries

7.4 Dashboard Features

The dashboard shall display:

- Domain inventory
- SSL expiry dates
- Status indicators
- Search and filtering
- Summary statistics
- AI-generated recommendations

7.5 Reporting

The system shall generate:

- CSV operational reports
- Markdown reports
- Incident summaries

8. Non-Functional Requirements

Requirement	Description
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Performance	Fast SSL scanning and analysis
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Reliability	Graceful failure handling
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Requirement	Description
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Scalability	Easy addition of new domains
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Maintainability	Modular architecture
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Security	No private key storage
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9. Technology Stack

Component	Technology
Programming Language	Python
Web Framework	Flask
Database	SQLite
AI Model	Ollama (Llama3)
Alerting	Discord Webhooks
Input Source	CSV
Reporting	CSV, Markdown
Version Control	GitHub

10. Workflow Plan

Phase 1 – SSL Monitoring

- Domain ingestion
- SSL certificate retrieval
- Expiry calculation

Phase 2 – Risk Engine

- Status classification
- Incident generation
- Risk evaluation

Phase 3 – AI Integration

- Ollama integration

- Renewal recommendation generation
- Priority assignment

Phase 4 – Dashboard Development

- Dashboard creation
- Search functionality
- Filtering features

Phase 5 – Reporting

- CSV export
- Markdown export
- Historical reporting

Phase 6 – Notification System

- Discord integration
- Alert generation

11. Business Rules

Rule 1

Every domain must be validated before SSL inspection.

Rule 2

SSL retrieval failures must be logged.

Rule 3

Domains with less than 30 days remaining are considered at risk.

Rule 4

Domains with less than 7 days remaining require immediate action.

Rule 5

AI-generated recommendations must be stored for auditing.

Rule 6

All monitoring activities must be recorded in SQLite.

Rule 7

Critical incidents must trigger Discord alerts.

Rule 8

Reports shall be generated after every monitoring cycle.

12. Expected Outputs

The completed system will:

- Monitor SSL certificates automatically
- Detect expiry risks
- Generate AI-based renewal recommendations
- Send real-time alerts
- Store monitoring history
- Export operational reports
- Display monitoring information via dashboard

13. Sample AI Recommendation

Task: Renew SSL certificate for wikipedia.org

Priority: High

Owner: Platform Team

Action Plan:

1. Generate CSR
2. Request certificate renewal
3. Install renewed certificate
4. Restart dependent services
5. Verify HTTPS functionality

14. Sample Discord Alert

SSL ALERT

Domain: wikipedia.org

Status: WARNING

Days Remaining: 27

Required Action: Renew SSL Certificate

15. Conclusion

The SSL Certificate Expiry Watcher provides an intelligent and automated approach to SSL lifecycle management.

The project demonstrates practical implementation of:

- AI Integration
- Incident Management
- Dashboard Development
- Real-Time Monitoring

By proactively detecting SSL risks and generating actionable recommendations, the system helps organizations improve reliability, security, and operational efficiency.